

Sultant Of Oman
Ministry of Health
Royal Hospital



سلطنة عمان
وزارة الصحة
المستشفى السلطاني

Document Management System



Sultanate of Oman
Ministry of Health
The Royal Hospital
Department of Engineering

Code: ENG-SOP-8-Vers.8

Effective Date: 14/01/2024

Target Review Date: 31/12/2026

Title: EQUIPMENT MANAGMENT PLAN

Purpose

To provide a safe, accessible and effective environment of care, consistent with the organization's mission, services and applicable governmental mandate.

The provision of safe and effective use of medical and engineering technology to minimize hazards and risks to patients, visitors, and staff.

To establish the framework, organization and processes for the development, implementation, maintenance, regulatory compliance, and continuous improvement of a comprehensive Medical and Engineering Equipment Technology Management Program.

Director Of Engineering Sections,

1. Biomedical Engineering Section.
2. Electrical / Electronic Section.
3. Mechanical Engineering Section.
4. Civil Engineering Section.
5. Air Condition Section.

1. Selecting and Acquiring of Medical and Other Engineering Equipment.

- 1.1. The hospital solicits input from individuals who operate and service equipment when it selects and acquires medical & engineering technology.
- 1.2. The hospital's medical and general stores department (Directorate of administrative affairs) and purchase department (Directorate of Finance department) is responsible for the purchase of all medical and engineering equipments.
- 1.3. The respective sections of engineering department (Biomedical / Mechanical/ Air conditioning / Civil) provides guidance and direction in the selection of medical & engineering equipment through active involvement in the capital process.
- 1.4. Respective engineering section also works with department managers to assist in the selection and purchase of medical and engineering equipment.
- 1.5. Respective engineering sections works with concern departments to provide average life expectancy, inventory, and information on equipment that has had extensive repairs or is no longer supported by the manufacturer.

2. Procedures for Inspection, Testing and Accepting of New Equipment.

- 2.1. The Al Shifa work request from medical and general store .
- 2.2. Equipment received from respective store with LPO copy, Service manual with spare part list and operational manual.
- 2.3. To carry out a electrical safety check.
- 2.4. Complete function check will be done.
- 2.5. If all the test passed then we stick a temporary Asset number.
- 2.6. The unit will return back to the medical / general equipment store with the copy of work request filled with the history of checking for acknowledgment.
- 2.7. The signed copy of work request along with documents will be filed for future reference.

3. Breakdown Of Equipment.

- 3.1. The Al Shifa breakdown work request from concern department.
- 3.2. The respective engineering section will act accordingly.
- 3.3. Technician / Engineer to attend the breakdown request in concern department.

3.5. Bring the equipment in the respective engineering section.

3.6. Repair the equipment if not required the spare parts and return to concern department after electrical safety and functional check.

3.7. Repair the equipment if requires spare parts , following are the three process of acquiring spare parts,

- Spare parts in-stock, Create a indent in Al Shifa system and collect it from engineering store.
- Spare parts not in stock, If normal purchase, The demand to purchase section.
- Float a tender and forward a comparative statement in director of engineering department for selection.
- The tender will send back to purchase department through administration department for issuing a LPO.
- Item received in engineering store and repaired the equipment.
- Spare parts not in stock, If emergency purchase, The quotation from the company maximum delivery 10 days after LPO.
- Forward the quotation to Asst. Director General (A&F) for approval through PE.
- Forward the approved documents to purchase department for issuing LPO.
- Item received in engineering store and repair the equipment.

3.8. After repair the breakdown work request will be signed by concern enduser and typed in Al Shifa.

3.9. The breakdown repair request will also be acknowledge by concern engineer from the related section of directorate of engineering and filed for future reference.

3.10. Work request will be close after entering the data in the Al-Shifa System.

4. Planed Preventive Maintenance (PPM).

4.1. The Al Shifa PPM work request will be generated in the system as per schedule.

4.2. The PPM list shared with the concern departments for arranging the equipments.

4.3. Technician / Engineer to attend the PPM request in concern department.

4.4. To check the history of equipment.

4.5. Bring the equipment in the respective engineering section.

4.6. To do the PPM of the equipment with manufacture defined checklist.

4.7. To carry out a electrical safety check.

4.8. Complete function check will be done.

- Spare parts in-stock, Create a indent in Al Shifa system and collect it from engineering store.
- Spare parts not in stock, If normal purchase, The demand to purchase section.
- Float a tender and forward a comparative statement in director of engineering department for selection and.
- The tender will send back to purchase department through administration department for issuing a LPO.
- Item received in engineering store and repaired the equipment.
- Spare parts not in stock, If emergency purchase, The quotation from the company maximum delivery 10 days after LPO.
- Forward the quotation to Asst. Director General (A&F) for approval through PE.
- Forward the approved documents to purchase department for issuing LPO.
- Item received in engineering store and repaired the equipment.

4.10. After repair the PPM work request the date will be entered in Al Shifa system.

4.11. Signed & finalize by related engineering section supervisor and engineer.

4.12. Filed the PPM request with checklist copy for future reference.

5. Incidents Related to Medical and Engineering Equipments

5.1. The hospital monitors and reports all incidents in which medical and engineering equipment is malfunctioning or attributed to serious injury, or serious illness of any individual.

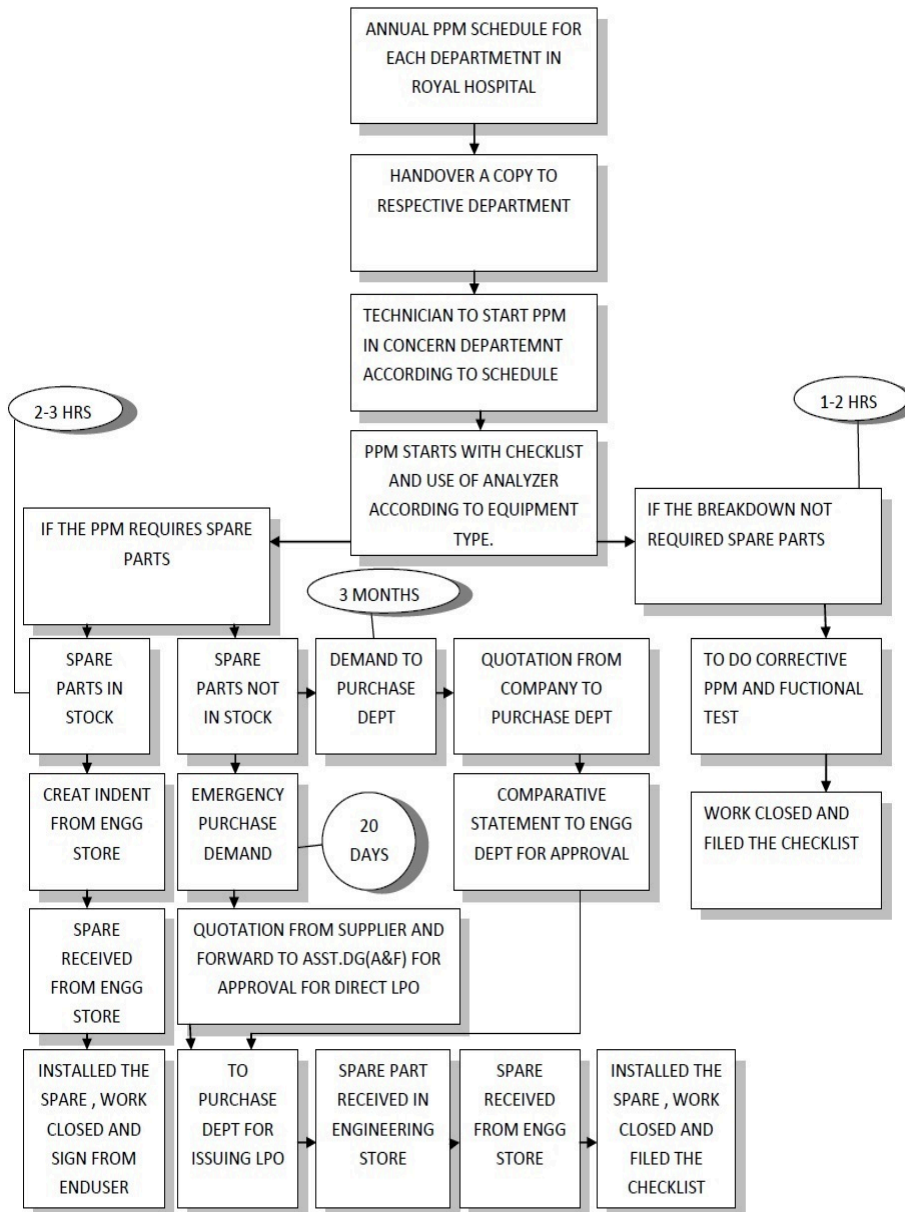
5.2. Information on hazard notices or recalls that apply to medical or engineering equipment are sent electronically to all managers of departments or patient units that use medical and engineering equipment.

5.3. Respective Engineering Section will follow all applicable policies and procedures, including checking the database to determine the presence of affected equipment and completing and documenting any appropriate work.

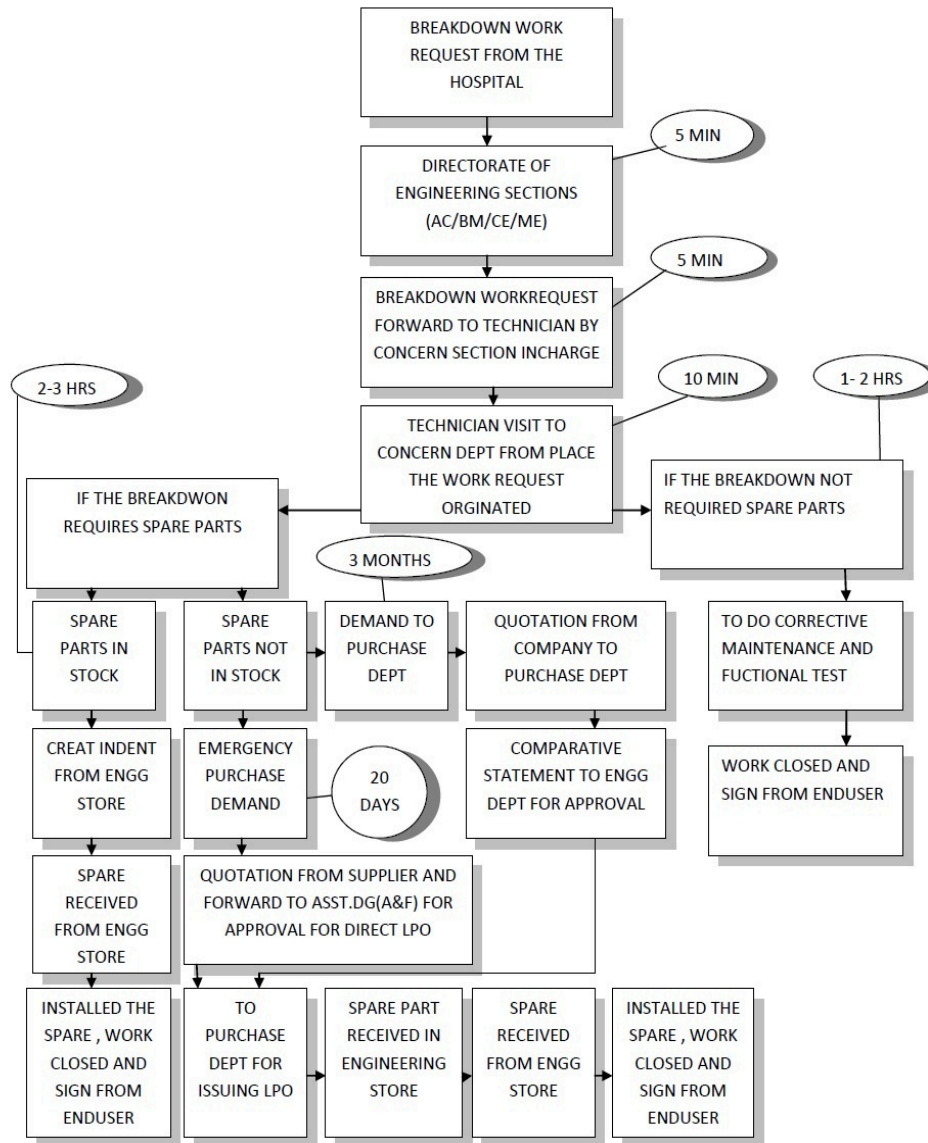
5.4. Respective Engineering shall assist risk management in conducting an investigation, evaluating the safety of the device, and determining whether the device should be impounded, repaired, further investigated, or returned to service. Safety related information should be communicated to the administration.

5.5. A summary of significant medical and engineering equipment incidents shall be reported to the Administration, HSE committee and Quality department.

PLANED PREVENTIVE MAINTENANCE FLOW CHART (BM/ME/CE/AC)



BREAKDOWN REQUEST FLOW CHART



Written By: Engr. Murtaza Muhammad Jaffar
Checked By: Engr. Adam Saif AlHudaifi
Authorized by: ADAM SAIF MOHD AL HADIFI

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Document Management System



Sultanate of Oman
Ministry of Health
The Royal Hospital
Department of Engineering

Code: ENG-SOP-9-Vers.9

Effective Date: 14/01/2024

Target Review Date: 31/12/2026

Title: Specialized Medical & Engineering Equipments.

Purpose

To provide a safe, accessible and effective environment of care, consistent with the organization's mission, services and applicable governmental mandate.

The provision of safe and effective maintenance of specialized medical and engineering equipments to minimize hazards and risks to patients.

All the specialized equipments are in contract with local authorized distributor of the product in order to maintain by trained product professional.

Breakdown of Specialized Equipments.

1. Request from Enduser.
2. Informed the local distributor engineer by enduser and concern engineer of the engineering section.
3. Response to breakdown as per contract with local supplier.
4. Completion of job as per contract with local supplier.

6. After completion of job the service report will be acknowledged by enduser, technician and concern engineers from engineering department.

Plan Preventive Maintenance (PPM) of Specialized Equipments.

1. Schedule of PPM by the authorized local agent of each contract equipment submitted at the beginning of each contractual year.
2. The PPM frequency planned according to the manufacture recommendation.
3. Equipment availability as per schedule for PPM after discussion with enduser and concern engineer from the engineering section.
4. After completion of PPM, the service report will be acknowledged by enduser, technician and concern engineers from engineering department.

Completion Certificate of Specialized Equipment for Payment.

1. Receive the invoice from the local authorized distributor in engineering department for the processing of payment.
2. Invoice will be sent to concern engineering section within the department.
3. Concern Engineer will make a completion certificate with comments and process through proper channel to finance through principal engineer for payment process.

Written By: Engr. Murtaza Muhammad Jaffar

Checked By: Engr. Adam Saif AlHudaifi

Authorized by: ADAM SAIF MOHD AL HADIFI

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Document Management System



Sultanate of Oman
Ministry of Health
The Royal Hospital
Department of Engineering

Code: ENG-SOP-2-Vers.1.0

Effective Date: 14/01/2024

Target Review Date: 31/12/2026

Title: Effectiveness of Preventive Maintenance

Purpose

To provide a safe, accessible and effective environment of care, consistent with the organization's mission, services and applicable governmental mandate.

The provision of safe and effective preventive maintenance of medical and engineering equipment to minimize hazards and risks to patients.

Following are the points:

1. Numbers of breakdown recorded at the time of Preventive maintenance to be verified, this with respect to the previous PPM.
2. Number of failure of equipment between consecutive PPMs. Lesser failure indicates possibility of effectiveness of PPMs.
3. Preventive maintenance to be done as per manufacturer recommendation and replace the wear & tear kits which are recommended in service manuals.
4. Technicians to perform the PPM as per manufacturer recommendation.
5. Test equipment used to perform PPM should have a valid calibration date and has to be periodically calibrated as per test equipment manufacture recommendation from accredited calibration center .

7. Recommended frequency of PPMs for medical and engineering equipments as per manufacturer.
8. Supervisor to randomly check the critical care equipments checklist and confirm the PPMs are done as per the manufacturer recommended checklist in the allotted time frame.
9. To ensure the availability of equipment for PPM, The list of equipment for PPM should be handed over to end-user in advance to arrange equipments for performing PPMs.
10. Regular interaction and feedback from enduser on equipment performance will help in improving the creditability of the work process.
11. Minimal spare parts usage on the equipment between consecutive PPMs indicates effectiveness of PPMs.

Written By: Engr. Murtaza Muhammad Jaffar

Checked By: Engr. Adam Saif AlHudaifi

Authorized by: ADAM SAIF MOHD AL HADIFI

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Information Technology Department'